

Introduction

Yerkes is known as the "birthplace of modern astrophysics." It is known for being home to the largest refracting telescope in the world, and a 180,000 historical glass plate collection.

The goal of this study is to see if there was a way to measure the expansion of planetary nebulae (PNe) by comparing historical observations to modern ones, despite their major differences.

In this study, I explored whether it was possible to see visible growth for planetary nebulae by comparing historical and modern observations, and, in doing so, developed the longest timeline of growth for the Dumbbell Nebula (M27) to date.

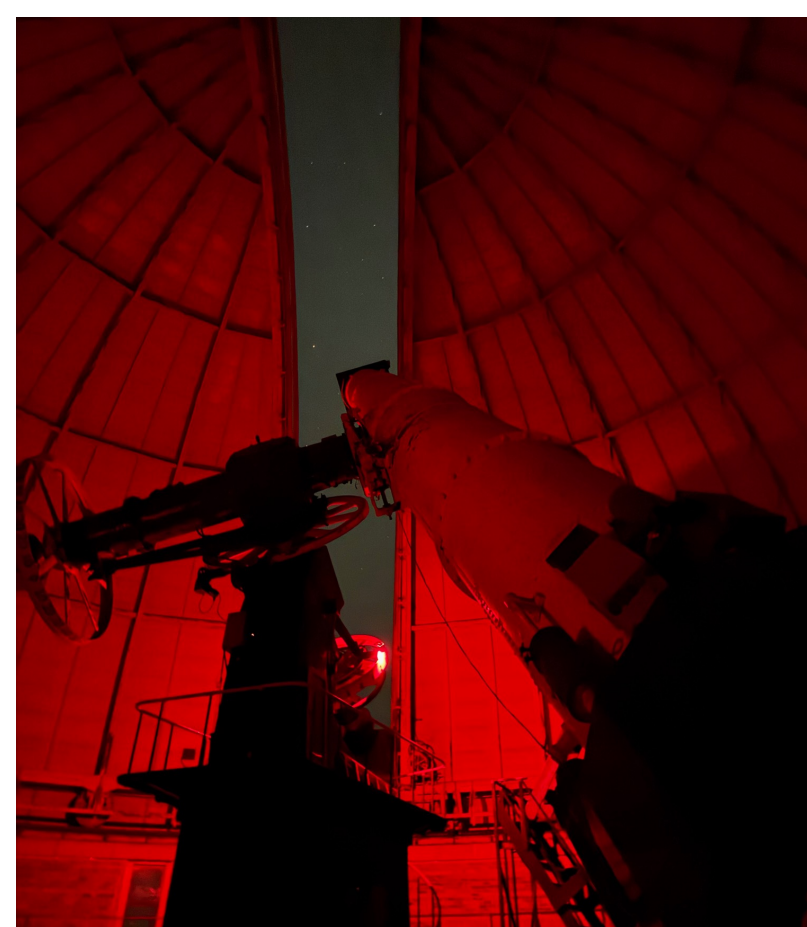


Figure 1: The Great Refractor.



Figure 2: A color composite image of the Dumbbell Nebula (M27), using g', i', r', H-alpha, and OIII filters. Made by me using Siril and GIMP.

Planetary Nebulae

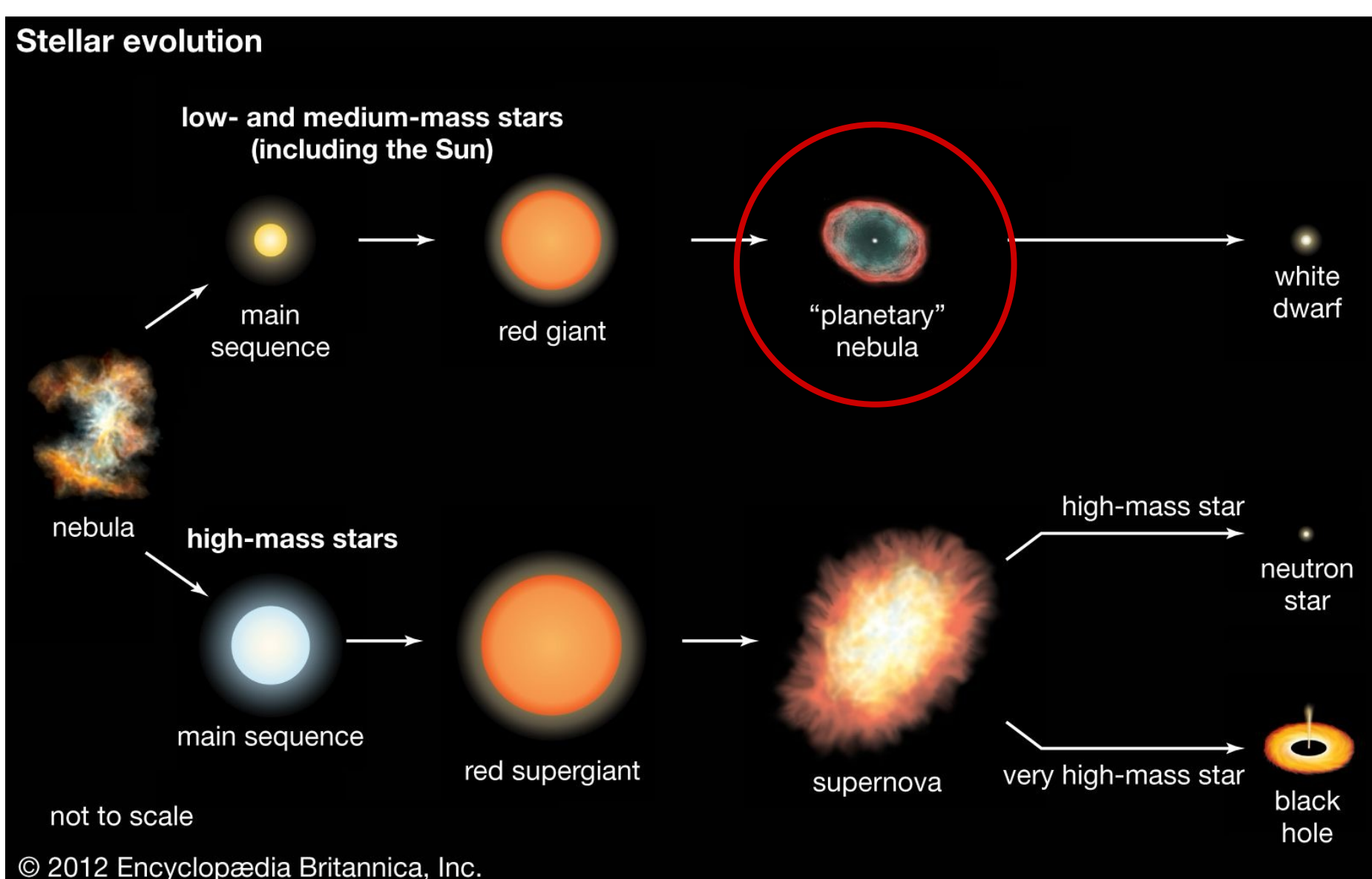


Figure 3: A visual of the stellar life cycle. Image credit to Encyclopaedia Britannica.

A planetary nebula (PN) is a stage in the stellar life-cycle.

PNe have a lifetime of ~10,000 years, shorter than the other stages of stellar life. This allows for changes to be observed on human timescales.

M27 is one of only four PNe included in the Messier Catalog and was the first planetary nebula ever discovered. This, and the abundance of quality observations of M27 in the Yerkes Plate Vault, made M27 an appealing target of study.

Observations

Modern images were taken on the Yerkes 41" Reflector (Y41), a 40" aperture telescope equipped with a complementary metal oxide semiconductor (CMOS) camera.

I conducted several observation runs over the course of the summer, using different filters (refer to Fig. 2). Only the empty-filtered images were used for scientific analysis.

I reduced all images with dark, flat, and bias frames. I also used the pinpoint astrometry program in MaximDL to assign coordinates to all the science frames, then stacked them to make a master image.



Figure 4: Observations in the Y41 with fellow interns!

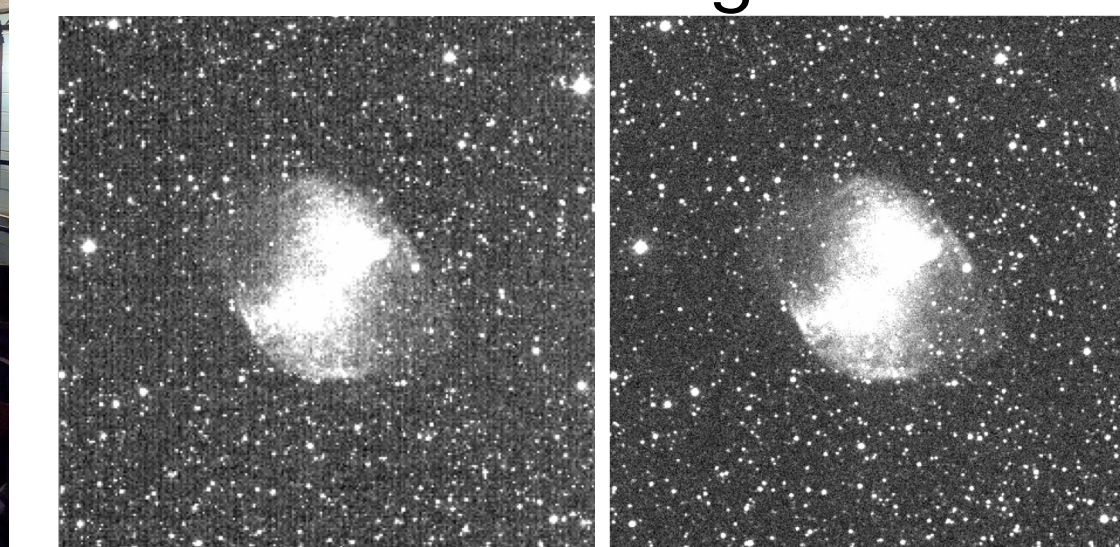
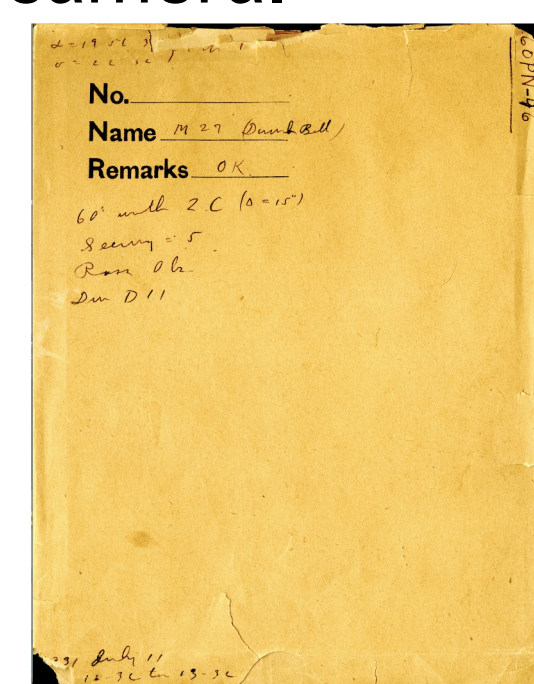


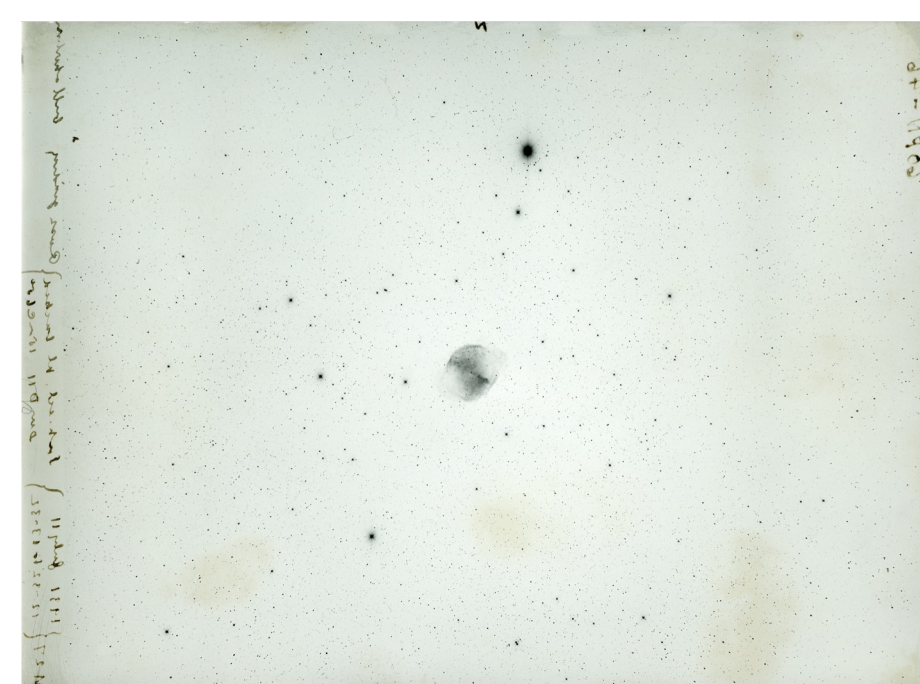
Figure 5: (a) - Unreduced science frame
(b) - Reduced science frame

Plate Digitization

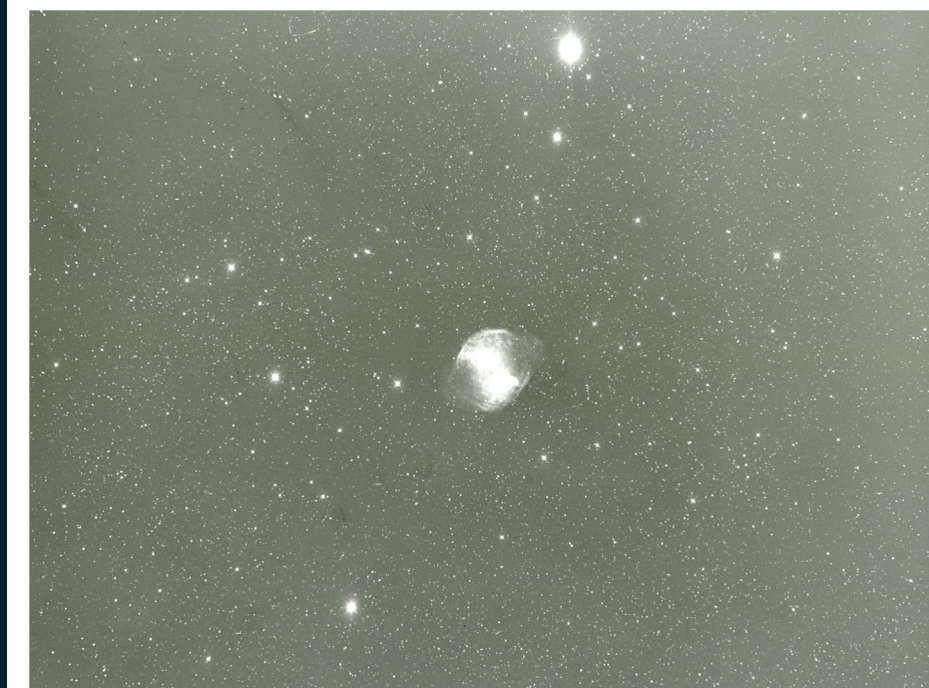
Plates must be scanned and digitized to allow for comparison with the CMOS images. Glass plates were popular for astronomical observations in the early 20th century. They capture images using light-sensitive emulsion (much like film cameras) with non-uniformly distributed photo sensitive grains that have lower saturation thresholds than the uniform pixel cells of a camera.



(a)



(b)



(c)



(d)

Figure 6: Plate 60PN-46, Mt. Wilson 1931

(a) - Envelope scan, 300 dpi
(b) - Negative full plate scan, 600 dpi
(c) - Positive center plate scan, 1600 dpi
(d) - Analyzed FITS image

Step 1: Find plates in the vault and perform scans to catalog the plate for future use.

Step 2: Convert the scan from a .TIFF to a .FITS (Flexible Image Transport System) File.

Step 3: Use nova.astrometry.net to insert the World Coordinate System (WCS) information into the FITS Header.

Step 4: Use the FITS Header information to find the pixel scale (arcsec/pixel) of the image.

Analysis

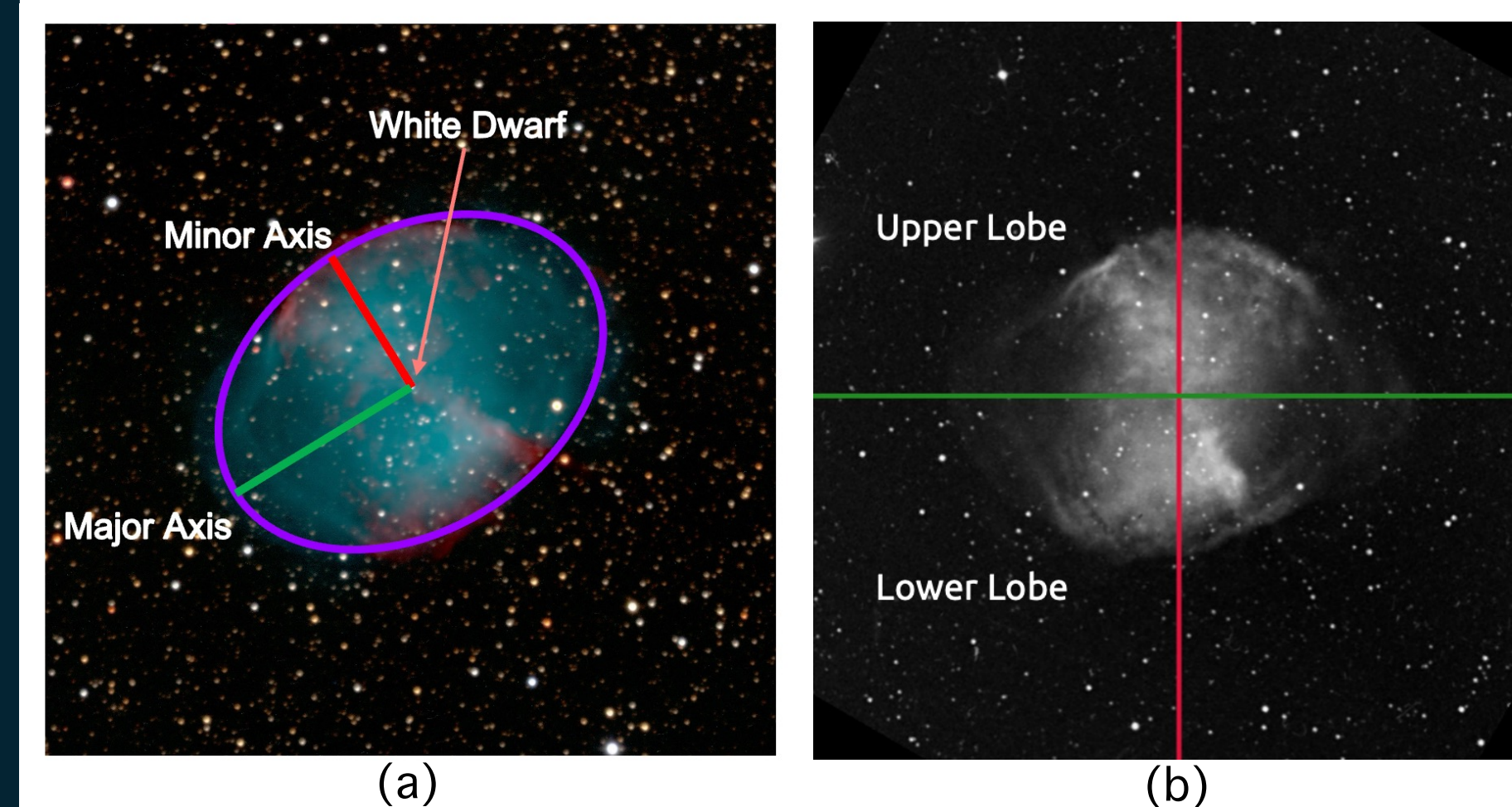


Figure 7: (a) - Ellipsoid shape of M27, major and minor axes marked
(b) - 30 degree rotation performed, upper and lower lobes marked

I cropped all images to be centered around the white dwarf and have dimensions of 13x13 arcmin. I then rotated the images by 30 degrees to make the minor axis vertical.

Across the minor axis, I used the pixel scale to average the centermost 5 arcsecs and extracted the pixel values of the image. This gives a curve proportional to flux. See Fig. 9 for an example of the flux curves.

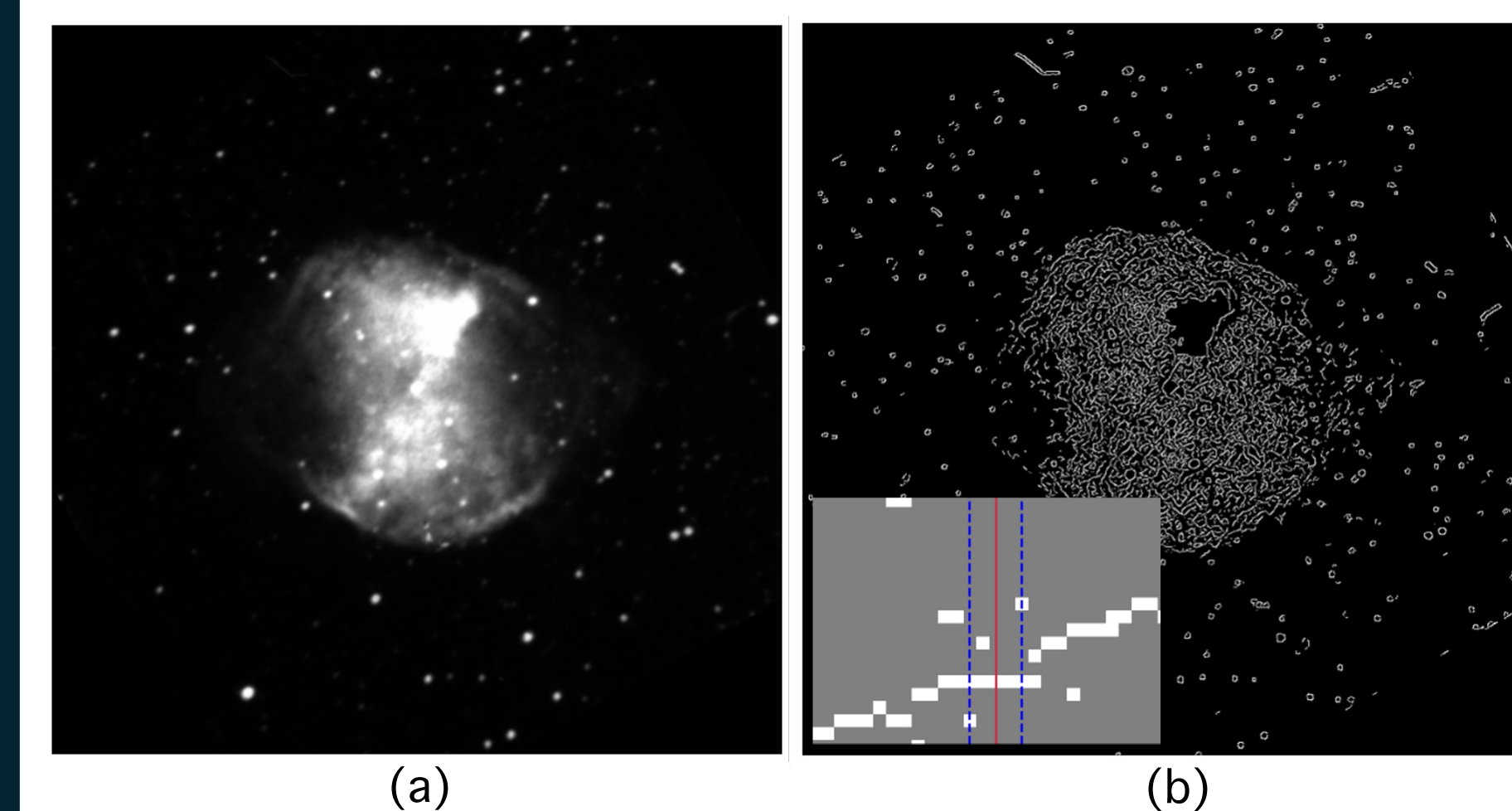


Figure 8: (a) - 1905 Plate
(b) - Extracted edges using Canny, zoom on edge.

I then used an edge-detection system called Canny to extract the edges of the nebula based on gradient intensity. I got a coordinate for the edge in pixels, then converted to arcseconds to allow for comparison between images of different resolutions.

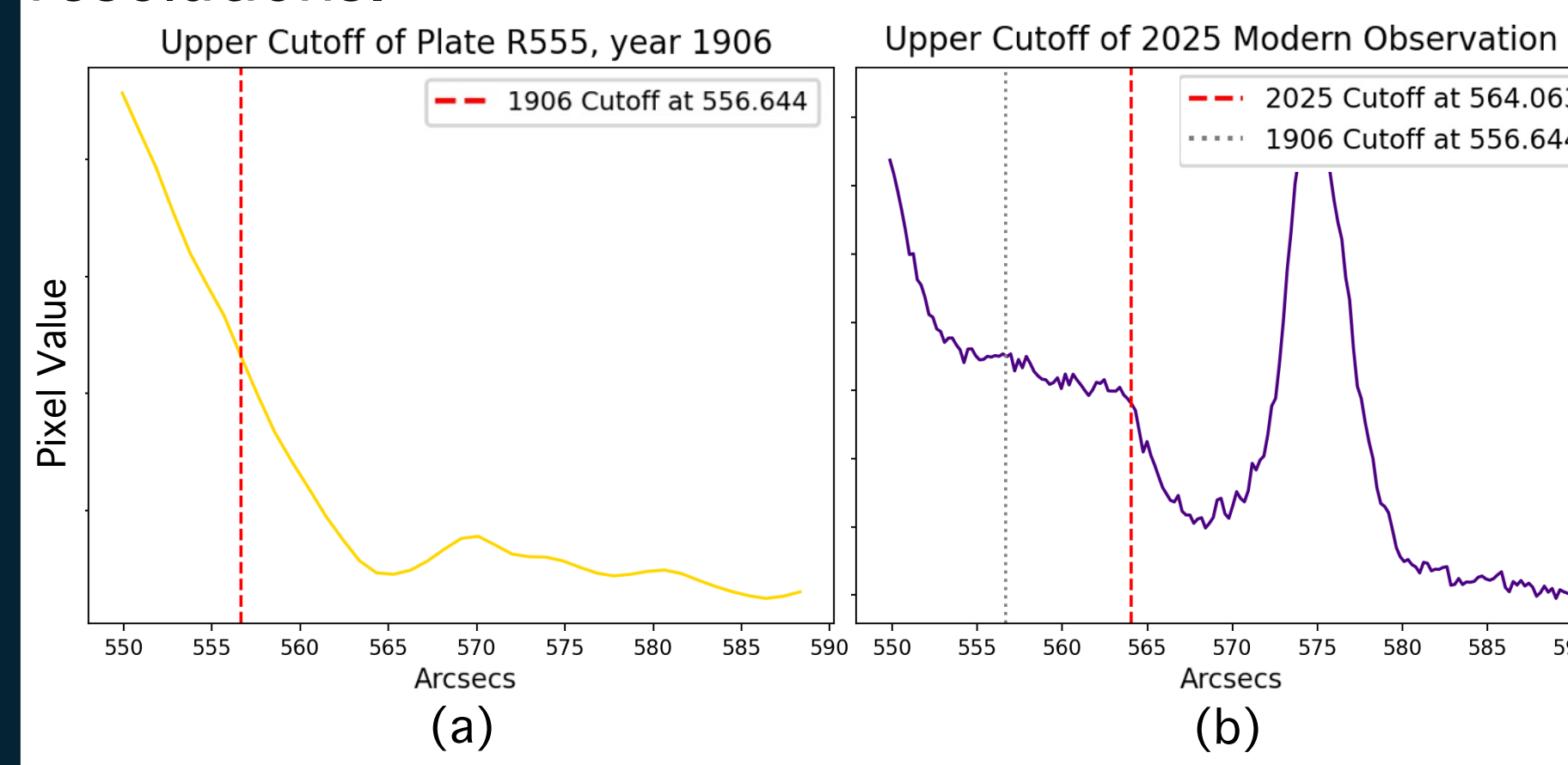


Figure 9: (a) - 1905 Upper Lobe Cutoff
(b) - 2025 Upper Lobe Cutoff

Subtracting the arcsec coordinates of the edges from the arcsec coordinate of the white dwarf allows me to find the radius of the nebula.

Acknowledgements

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References

Distance from White Dwarf Over Time

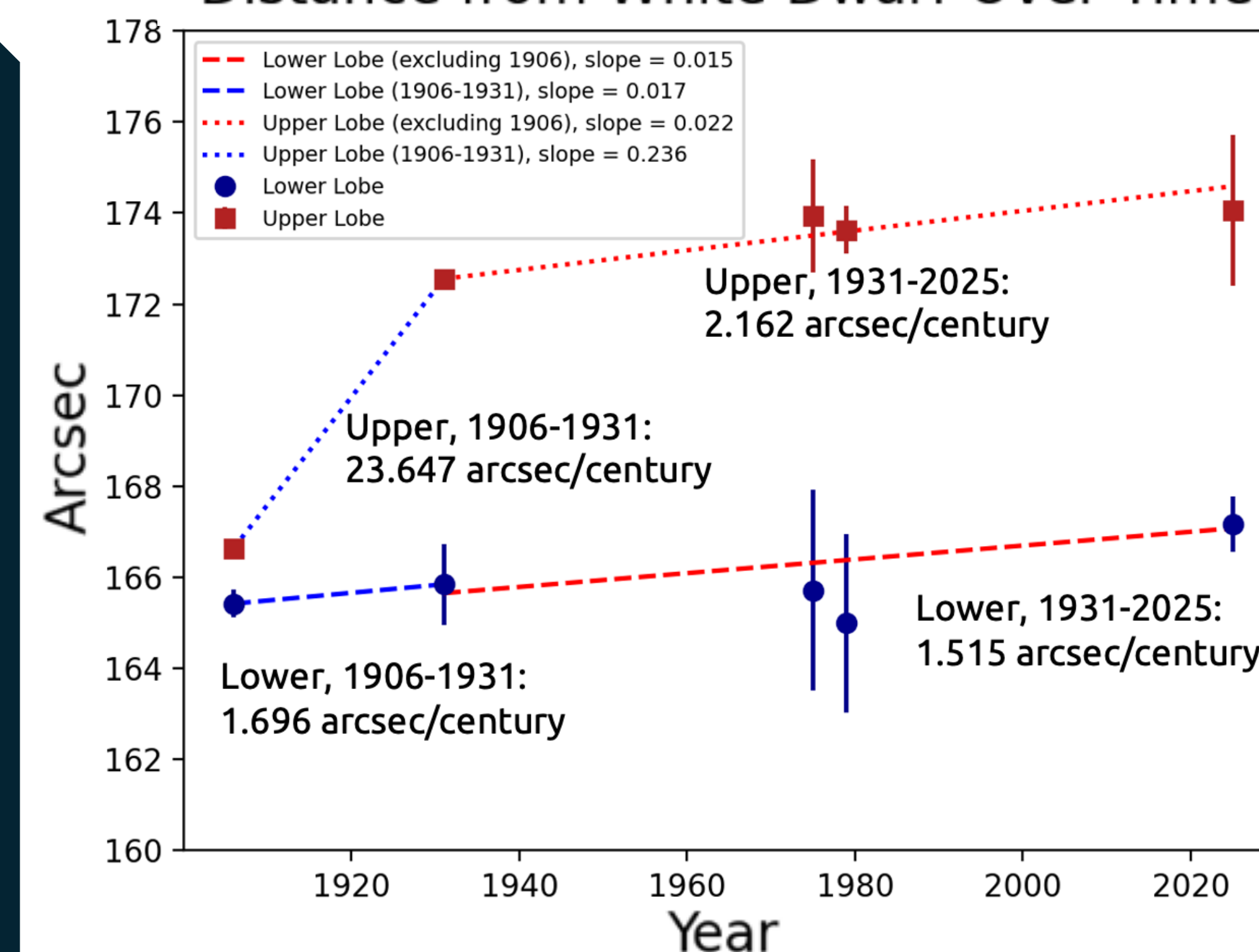


Figure 10: Growth of M27, separated by upper and lower lobe.

Results and Discussion

We see an average radial growth of 1.839 arcsecs per century, when only considering observations from 1931 onwards.

- The Yerkes Observatory plate vault can be used for meaningful analysis of nearby PNe along with modern observations.
- The period 1931-2025 shows radial growth consistent with published values for M27 (~2 arcsec/century).
- The upper lobe of M27 appears to be growing at a faster rate than the lower lobe in the modern era.
- There's a jump in growth of the upper lobe in the 1906-1931 era, likely due to differences in color sensitivity between earlier and later emulsion types.

Possibilities for the future of this study are:

- Researching the emulsion types used for each plate in the study, as earlier plates had emulsions that were more blue-sensitive, while later emulsions were more red-sensitive.
- Performing a relative flux calibration to evaluate the low-surface-brightness areas of the nebula over time.

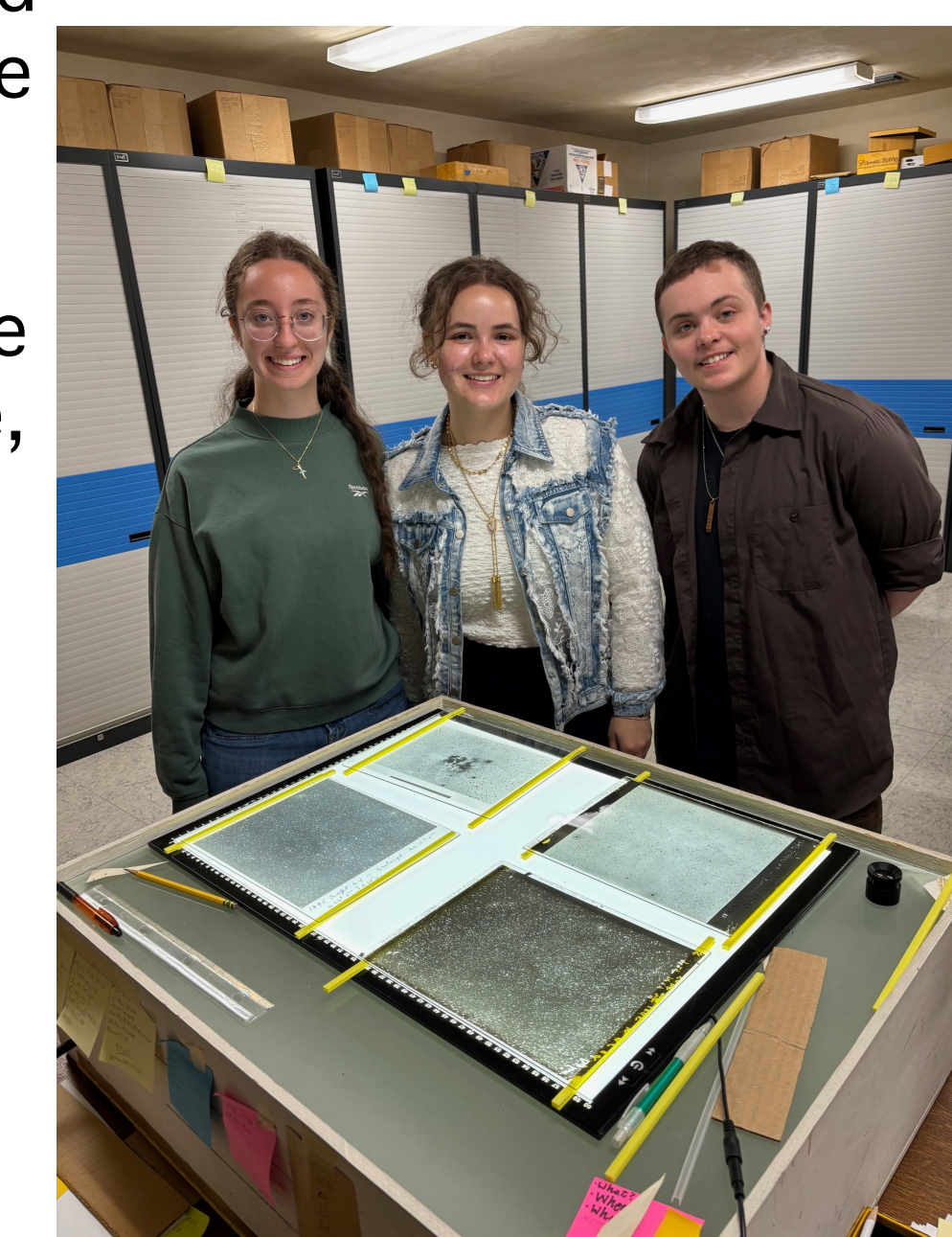


Figure 11: The Yerkes Plate Vault with fellow interns!